

4. The Hawaiian archipelago is a group of tropical volcanic islands that formed within the last 30 million years. They are located in the middle of the Pacific Ocean, almost 2300 miles away from the closest continent, North America. There are 351 different bird species on the islands, of which 59 species are endemic (found only on the Hawaiian Islands). The UK has 635 different bird species, of which only one species is endemic.
- (a) (i) Calculate the percentage of bird species that are endemic to Hawaii. [1]
- %
- (ii) Less than 0.2% of UK bird species are endemic. Suggest **one** reason for the difference in the percentage of endemic species between the UK and Hawaii. [1]
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- (b) Hawaiian honeycreepers are a group of birds endemic to Hawaii. At least 56 species are known to have existed, however 38 species are now extinct. **Image 4.1** and **Table 4.2** show one species, called apapane (*Himatione sanguinea*), and part of its classification.
- Complete Table 4.2**, to show the full classification of apapane. [3]

Image 4.1



Table 4.2

Domain	
Kingdom	Animalia
Phylum	Chordata
Class	Aves
.....	Passeriformes
.....	Fringillidae
Genus	
Species	



- (c) Scientists have debated the identity of the ancestor of the honeycreepers. Morphological and behavioural data have suggested several possibilities. One group of scientists carried out DNA hybridisation to compare the DNA of apapane with other bird species. They include the orchard oriole (*Icterus spurius*), the palm tanager (*Thraupis palmarum*) and the purple finch (*Carpodacus purpureus*). This technique involves:
- Heating the extracted DNA from apapane to break the hydrogen bonds, causing the two strands to separate.
 - Mixing the separated strands with single DNA strands from a different species.
 - Cooling the mixture to allow hydrogen bonds to form between complementary base pairs to produce hybrid DNA (with one strand from apapane and the other strand from the different species).

When the hybrid DNA is then reheated the strands will separate at a lower temperature. This is because fewer hydrogen bonds will be present than in non-hybrid DNA from apapane. The difference in temperature indicates how similar the DNA sequences are.

The results are shown in **Table 4.3**:

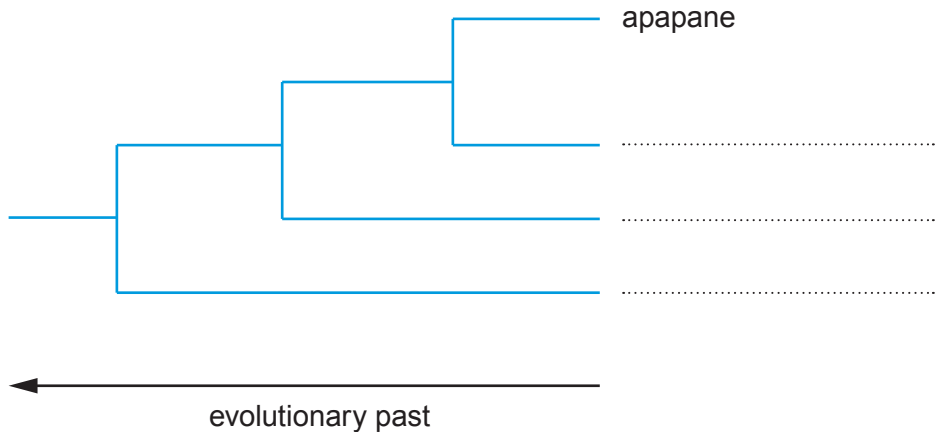
Table 4.3

DNA	Temperature required to separate hybrid strands / °C
apapane – apapane	92.0
apapane – oriole	84.9
apapane – tanager	84.5
apapane – finch	87.8

- (i) Use the results to **complete the phylogenetic tree in Image 4.4**.

[2]

Image 4.4



Examiner
only

(ii) Use the information provided to explain how the technique allows the evolutionary relationships between the species to be determined. [3]

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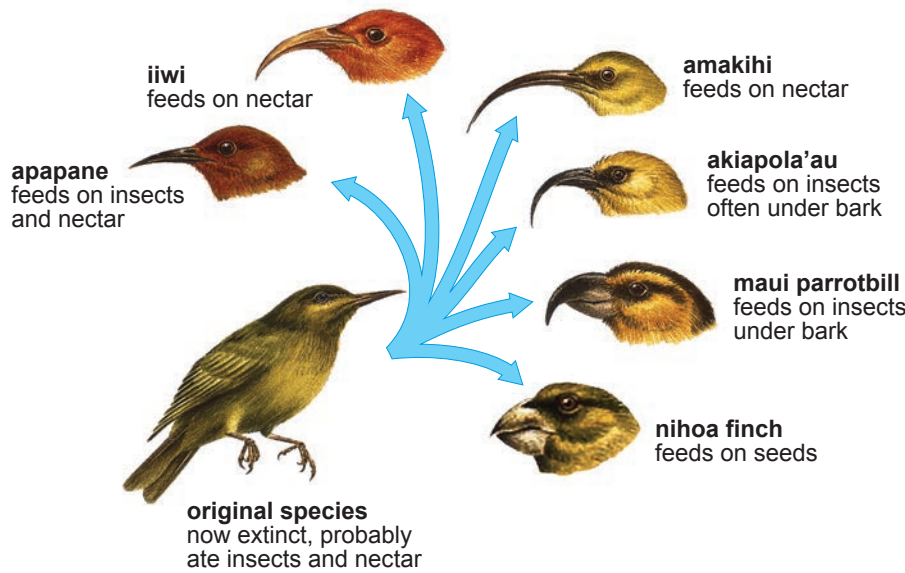
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- (d) Since the arrival of the common ancestor in Hawaii, the honeycreepers have diversified into many different species. **Image 4.5** shows some of these with information regarding their food source.

Image 4.5



- (i) The apapane and iiwi are found in similar habitats and have similar diets. Explain why they are regarded as separate species. [1]

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- (ii) Using the information provided, describe and explain the evolutionary change illustrated by **Image 4.5**. [4]

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